

Algorithmic Trading: Automating Financial Investment Decisions

Algorithms can help automate trading decisions using financial theory, statistical modelling, and computer programming. Python has emerged as a popular language for algorithmic trading.



The use of computer algorithms to automatically carry out trading decisions in financial markets is known as algorithmic trading, or ‘algo trading’. Traders create mathematical models and rules that control when and how trades should take place rather than manually giving buy or sell orders. To find profitable trading opportunities, these algorithms examine market data, including price, volume, volatility, and other indicators.

In today’s financial markets, algorithmic trading has taken centre stage. To execute strategies quickly, accurately, and consistently, large institutional investors, hedge funds, and proprietary trading firms mainly rely on automated trading systems. The adoption of algorithmic trading has been further expedited by the quick development of computer technologies, high-speed networks, and data availability.

Python has become one of the most widely used programming languages for algorithmic trading because of its ease of use, adaptability, and plenty of libraries for financial modelling, machine learning, and data analysis.

Concept of algorithmic trading

In financial markets, algorithmic trading is the process of creating trading signals and carrying out orders automatically using pre-established rules and computing models.

An algorithmic trading model’s general structure can be expressed as:

Trading signal = f (Market data, Indicators, Model parameters)

...where:

- Market data includes price, volume, and other financial variables.
- Indicators represent derived metrics such as moving averages or momentum.
- Model parameters are predefined thresholds or coefficients.

An automatic execution engine places the order in the market after a trading signal is issued.